Project Charter

Embeddable Cross-Course AI Agent General Architecture Platform

1. Project Identification

Field	Details
Project Name	Embeddable Cross-Course AI Agent General Architecture Platform
Date of Authorization	April 12, 2026
Project Start Date	March 30, 2026
Project End Date	June 14, 2026
Version	1.1

2. Project Stakeholders

Name	Title / Role	Student ID	E-mail
冯俊财	AI Architect / Backend	2353924	20614667@qq.com
路清怡	Backend	2352996	849014041@qq.com
张诗蔻	Frontend	2353240	2987991635@qq.com
纪鹏	Test / Frontend	2351869	jp3274313334@163.com

3. Project Description

Background:

The current education sector faces the phenomenon of "fragmented intelligent teaching." Specifically, various courses develop independent intelligent teaching systems, leading to architectural heterogeneity and duplicate construction due to the lack of technical standardization. These systems are difficult to seamlessly embed into mainstream teaching platforms like Chaoxing and DingTalk, causing integration difficulties. Furthermore, course-specific knowledge is hard to transfer effectively, resulting in fragmented knowledge

presentation. Existing systems also provide rough teaching guidance, lack refined analysis and contextual annotation for assignment details.

Description of the challenge or opportunity:
There is a pressing need for a standardized, reusable AI Agent architecture that enables different courses to rapidly build and deploy intelligent teaching assistants without redundant engineering effort. Large Language Model (LLM) technology now makes it feasible to provide annotated, context-aware feedback at scale—transforming AI grading from "usable" to "precisely useful."

Overview of the desired impact:
This project delivers EduAI, a general-purpose AI Agent platform that: (1) provides a standardized SDK for rapid course-Agent development; (2) enables seamless embedding into Chaoxing and DingTalk via a lightweight iframe Widget; (3) performs annotation-style assignment grading with ⩾ 90% accuracy; (4) analyzes student learning states and generates targeted exercises to close individual knowledge gaps.

4. Measurable Organizational Value

Primary MOV: Enable universities to reduce the cost and time of building intelligent teaching systems for new courses, while delivering grading feedback quality that matches or exceeds manual teacher annotation.

Area	Goal	Metric	Target Value
Quality	Deliver accurate in-course Q&A	Answer accuracy on core knowledge points	⩾ 90%
Quality	Deliver teacher-level grading feedback	Key error annotation accuracy (location + content)	⩾ 90%
Cost	Reduce duplicate AI system development effort	Reuse one SDK for ⩾ 2 course types without code changes	Demonstrated in demo
Performance	Support large-scale concurrent access	Concurrent users supported	⩾ 500
Adoption	Enable embedding in mainstream LMS platforms	Platforms supported via iframe/Widget	⩾ 2 (Chaoxing, DingTalk)

5. Project Scope

What is included in the scope of this project:

1. Unified AI Agent SDK (AgentBase,QAAgent,GradingAgent, ExerciseAgent) supporting multiple course types
2. Knowledge middleware: document ingestion (PDF/Word/PPT), vector embedding (pgvector), knowledge graph, hybrid retrieval (vector + BM25 + Rerank)
3. Refined intelligent engine: annotation-style grading (text, code, image), learning analytics (mastery matrix, early warning), personalized exercise generation
4. RESTful API backend (FastAPI, 8 route groups: Auth, Courses, Agents, Assignments, Chat, Analytics, Exercises, Platform)
5. Full frontend management portal (React 18, 9 pages): Login, Dashboard, Course Management, Intelligent Q&A, Assignment Management, Grading Result Viewer, Analytics Dashboard, Exercise Center, Agent Visual Builder
6. Embeddable Widget (iframe-compatible lightweight chat/grading window for LMS integration)
7. Platform adaptation layer: technical specification and simulated interfaces for Chaoxing (LTI) and DingTalk (H5)
8. Cloud deployment: Docker Compose + Nginx + HTTPS on a cloud server

What is outside the scope of this project:

1. Native mobile application (iOS/Android)
2. Real-time live-class features (video streaming, screen sharing)
3. Full production API certification and live integration with Chaoxing or DingTalk (simulated adaptation only, as permitted by competition rules)
4. Automated billing or payment systems
5. Support for physical lab equipment or hardware-linked experiments

6. Project Schedule Summary

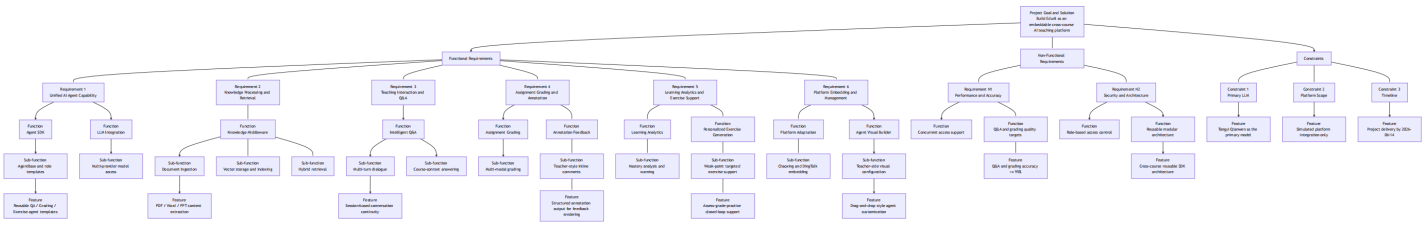
- **Project start date:** March 30, 2026
- **Project end date:** June 14, 2026
- **Duration:** 11 weeks

Key Milestones:

Milestone	Description	Target Date
M1	Foundation Ready: environment running; LLM API connected; database schema migrated; JWT auth verified; Celery/Redis and MinIO validated	April 12, 2026

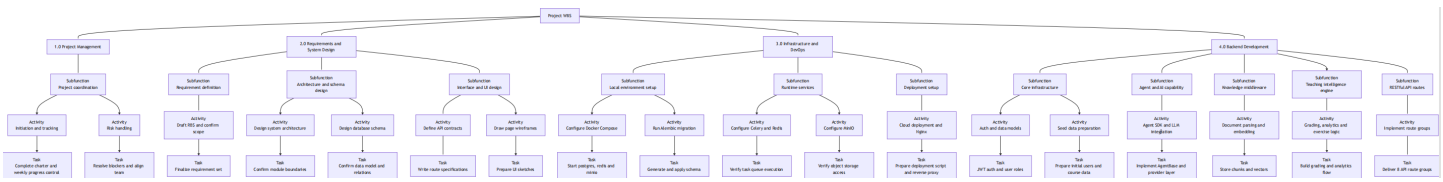
M2	Q&A MVP: course material upload, resource processing, and course-aware AI Q&A available (accuracy ⩾ 90%)	April 26, 2026
M3	Full teaching loop: submit assignment → AI annotates → learning analytics → exercises generated	May 17, 2026
M4	Full feature integration: platform embedding and Agent visual builder completed; Q&A ⩾ 90%; grading ⩾ 90%	May 31, 2026
M5	System fully deployed on cloud server; end-to-end demo validated; project documentation complete	June 14, 2026

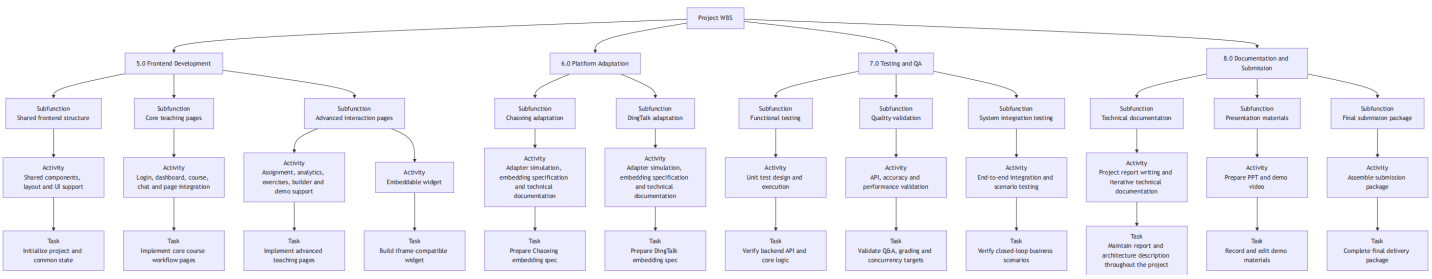
7. Requirement Breakdown Structure (RBS)



8. Work Breakdown Structure (WBS) & Hours Estimation

To better display the WBS, we put it in two diagraphs.





WBS	Module	Estimated Hours	Primary Responsible
1	Project Management	24	All members
2	Requirements and System Design	40	All members
3	Infrastructure and DevOps	32	冯俊财 / 路清怡
4	Backend Development	120	冯俊财 / 路清怡
5	Frontend Development	96	张诗蔻 / 纪鹏
6	Platform Adaptation	16	纪鹏 / 路清怡
7	Testing and QA	40	纪鹏 / All members
8	Documentation and Submission	32	纪鹏 / All members
	Total Working Hours	400	

WBS	Task / Work Package	Estimated Hours	Suggested Owner
1.1	Complete charter and weekly progress control	16	All members
1.2	Resolve blockers and align team	8	All members
2.1	Finalize requirement set	8	All members
2.2	Confirm module boundaries	12	All members
2.3	Confirm data model and relations	8	冯俊财 / 路清怡
2.4	Write route specifications	8	冯俊财 / 路清怡
2.5	Prepare UI sketches	4	张诗蔻
3.1	Start postgres, redis and minio	8	冯俊财
3.2	Generate and apply schema	4	冯俊财
3.3	Verify task queue execution	8	路清怡
3.4	Verify object storage access	4	路清怡
3.5	Prepare deployment script and reverse proxy	8	冯俊财 / 路清怡
4.1	JWT auth and user roles	10	冯俊财
4.2	Prepare initial users and course data	10	路清怡
4.3	Implement AgentBase and provider layer	16	冯俊财
4.4	Implement QA / grading / exercise agent	16	冯俊财

	capability		
4.5	Store chunks and vectors	12	路清怡
4.6	Build grading and analytics flow	16	冯俊财 / 路清怡
4.7	Deliver Auth / Courses APIs	6	路清怡
4.8	Deliver Agents / Assignments APIs	6	冯俊财 / 路清怡
4.9	Deliver Chat / Analytics APIs	6	冯俊财 / 路清怡
4.1	Deliver Exercises / Platform APIs	6	路清怡
4.11	Resource upload and async processing loop	8	路清怡
4.12	Q&A closed-loop integration	8	冯俊财
5.1	Initialize project and common state	8	张诗蔻 / 纪鹏
5.2	Implement login, dashboard, course and chat pages	32	张诗蔻 / 纪鹏
5.3	Implement assignment, analytics, exercises and builder pages	40	张诗蔻
5.4	Build iframe-compatible widget	16	张诗蔻 / 纪鹏
6.1	Prepare Chaoxing embedding spec	8	纪鹏
6.2	Prepare DingTalk embedding spec	8	纪鹏
7.1	Unit test design and execution for backend/API logic	14	纪鹏
7.2	API, Q&A, grading and concurrency validation	14	纪鹏
7.3	End-to-end integration and closed-loop scenario verification	12	All members
8.1	Project report writing and iterative update	12	All members
8.2	Record and edit demo materials	8	纪鹏
8.3	Complete final delivery package	12	纪鹏

9. Project Budget Summary

Item	Description	Estimated Cost (CNY)
Cloud Server	4 vCPU / 8 GB RAM, rented as needed for deployment	¥ 200
LLM API Credits	Tongyi Qianwen (qwen-max / qwen-vl-max)	¥ 300
Object Storage	Aliyun OSS / MinIO for course files	¥ 200

Total		¥ 700

10. Quality Issues

Specific quality requirements:

Requirement	Metric	Target
Q&A accuracy	Core knowledge-point answer correctness	⩾ 90%
Grading annotation accuracy	Key error annotation (location + content) correctness on sample assignments	⩾ 90%
System availability	Uptime during demo and evaluation period	⩾ 99%
API response time	Non-AI endpoint latency	< 2 seconds
Concurrency	Simultaneous active users	⩾ 500

Quality assurance approach:

1. All backend pull requests require peer code review before merging.
2. Q&A accuracy is measured against a 50-question ground-truth set per course.
3. Grading accuracy is measured against the sample assignments provided by the competition (key-error annotations pre-labeled by team members).
4. Celery async tasks and Redis caching are used to maintain responsiveness under load.

11. Resources Required

People:

4 team members (roles defined in Section 2)

Technology:

Category	Tool / Technology
Frontend	React 18, TypeScript, Vite, Ant Design 5, ECharts, React Flow
Backend	FastAPI (Python 3.11), SQLAlchemy 2.0, Alembic, Celery
AI / LLM	LangChain 0.3, LangGraph, Tongyi Qianwen (qwen-max, qwen-vl-max), DeepSeek
Database	PostgreSQL 16 + pgvector, Redis 7, MinIO

DevOps	Docker Compose, Nginx, Aliyun ECS, GitHub Actions
Collaboration	GitHub (code), Feishu (communication), VS Code

Facilities:

1. Cloud server (Aliyun ECS, provisioned by Week 2)
2. Each member's local development machine
3. GitHub repository for version control and CI

12. Assumptions and Risks

Assumptions:

1. The team has valid API access to Tongyi Qianwen (qwen-max, qwen-vl-max) throughout the project.
2. Competition rules allow simulated platform adaptation; real Chaoxing/DingTalk API certification is not required.
3. All team members can commit approximately 10 hours per week to the project.

Key Risks:

Risk	Probability	Impact	Mitigation Strategy
LLM API latency degrades user experience	Medium	High	SSE streaming output; Redis caching for frequent questions
Grading / Q&A accuracy fails to reach 90%	Medium	High	Per-type prompt templates; few-shot examples; iterative calibration with sample data
Schedule slippage due to exam season (late May)	Medium	Medium	Front-load core AI features in Weeks 3–7; documentation and video in final 2 weeks
Insufficient demo data for convincing showcase	Low	Medium	Team member dedicates time in Weeks 1–4 to prepare 2–3 complete course datasets
Chaoxing/DingTalk real integration complexity	Low	Low	Competition rules permit technical-spec-only adaptation; no live integration required

Constraints:

1. Fixed end date: June 14, 2026; no extensions possible.

2. Budget capped at approximately ¥ 800 (cloud + LLM API costs).
3. Team size is fixed at 4 members; no external contractors.

13. Project Administration

Communications Plan:

1. Weekly team sync meeting to review progress against milestones.
2. Daily async updates via group chat (Feishu) for blockers and code review requests.
3. GitHub Issues used to track bugs; GitHub Projects board tracks WBS task status.
4. All code merged to the main branch via Pull Requests only.

Scope Management Plan:

1. Any proposed scope changes must be evaluated for impact on the June 14 deadline before acceptance.
2. Core features (Q&A, grading, analytics, exercises) take priority. Creative features (Agent visual builder, platform adapter) are secondary if time is short.
3. The Project Manager has final authority on scope decisions.

Quality Management Plan:

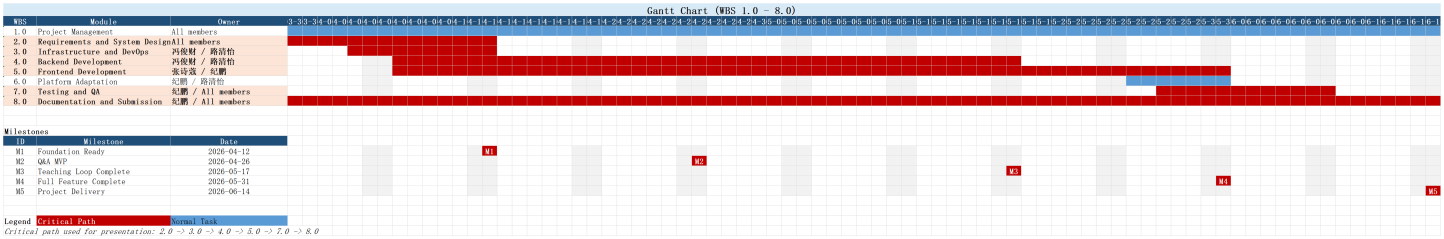
1. Mandatory peer code review for all pull requests before merging.
2. Q&A and grading accuracy tested at each milestone; results recorded and prompt templates revised if targets are missed.
3. End-to-end integration test performed in Week 10 covering all 9 UI pages and all 8 API route groups.

14. Acceptance and Approval

This Project Charter is approved by all team members, confirming the project objectives, scope, schedule, and responsibilities described above.

Role	Name	Signature	Date
Team Leader	冯俊财	冯俊财	April 12, 2026
Team Member	张诗蔻	张诗蔻	April 12, 2026
Team Member	纪鹏		April 12, 2026

		纪鹏	
Team Member	路清怡	路清怡	April 12, 2026

15. Appendices

Appendix A — Gantt Chart